GIS & Remote Sensing Training Plan for Rainfall Network Design

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Inhaltsverzeichnis

Goal

This modular training program is designed for a seminar on physiographically-stratified rainfall networks in low mountain regions (e.g., Burgwald, Germany). It uses the R-spatial ecosystem and provides a mix of core and elective modules tailored to group projects. Each session is 3 hours long, with a maximum of 1 hour of instructor-led input.

Structure

- **Total sessions:** roughly 10 × 3h
- **Core modules (required):** 6 modules, spread over sessions 1–6
- **Advanced (toolkit) modules:** Optional, to support specific project needs, scheduled flexibly across sessions 7–10

Core Modules (for all groups)

Module 1: Project Organization & FAIR Data Retrieval

- R project setup: `renv`, `here`, `targets`
- FAIR principles (Findable, Accessible, Interoperable, Reusable)
- Open datasets: DGM (elevation), CORINE/ATKIS (land use), RADOLAN, Sentinel-2, stream gauge data

Module 2: Geodata Preprocessing

- Raster and vector import, reprojection, masking (`terra`, `sf`, `stars`)
- Deriving morphometric indices: slope, aspect, convexity, TWI
- Simplifying land cover classes for analysis

Module 3: Interpolation Techniques

- Deterministic: IDW, Spline, trend surfaces
- Geostatistical: Ordinary/Universal Kriging, CoKriging (elevation as covariate)
- Parametric: Regression models based on physiographic predictors (PRISM-style)

Module 4: Wind Fields & Luv-Lee Classification

- Wind data (DWD stations, ERA5/COSMO reanalysis)
- Deriving dominant wind directions (seasonal, annual)
- Linking wind data with aspect for Luv/Lee mapping

Module 5: Spatial Structuring & Stratification

- **Goal:** Segment study area into physiographically consistent zones
- Clustering: `kmeans`, `raster.kmeans`
- Nowosad-style landscape pattern analysis:
  - `supercells`: superpixel-based segmentation
  - `motif`: landscape pattern frequency analysis
  - `spQuery`: neighborhood similarity queries
  - `patternogram`: spatial pattern comparison
- Sampling representative locations (`spatSample`, `sf::st_sample()`)

Module 6: Visualization & Mapping

- Mapping with `tmap`, `mapview`, `ggplot2`
- Composite visual layers: terrain + stations + uncertainty + structure

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Advanced Toolkit Modules (optional per group/project)

For Module 1 (Project organization):

- Metadata management with YAML, `EML`
- Project transparency with GitHub workflows

For Module 2 (Preprocessing):

- Landform classification (ridge/valley/slope)
- Watershed and flow direction (`terra`, `whitebox`, `RSAGA`)

For Module 3 (Interpolation):

- Radar-gauge bias correction (mean field bias, rolling bias)
- Time series smoothing, rolling event detection (`zoo`, `hydroGOF`)

For Module 4 (Wind):

- Automated Luv/Lee classification per aspect and wind dominance
- Wind-exposure stratification zones

For Module 5 (Structuring):

- Cluster validation (silhouette score, within-cluster variance)
- Comparison: PRISM/ASOADeK-type models vs. unsupervised clustering

For Module 6 (Visualization):

- Optional: interactive dashboards with `shiny`, `leaflet`

Suggested Timeline (Flexible)

Session	Theme	Module(s)	Format
1	Project & Data Infrastructure	Module 1	Plenary + group setup
2	Data Preprocessing	Module 2	Guided workflow + maps
3	Structuring & Patterns	Module 3	Core + Nowosad extensions
4	Wind Fields & Luv/Lee	Module 4	Analysis + map output
5	Interpolation Basics	Module 5	Demo + application
6	Visualization	Module 6	Output & comparison
7–10	Workshops	Advanced modules	Project-specific tutorials

This structure ensures every group acquires key spatial methods while allowing targeted deep-dives into advanced techniques, including the use of cutting-edge spatial pattern tools from the Nowosad ecosystem.